

Nawapan (Ploy) Wattanawanichkul

ACADEMIC APPOINTMENTS

2026–Present **University of Virginia (UVA)** VA, USA
Whyburn Research Associate and Lecturer
Mentor: Peter Humphries

EDUCATION

2021–2026 **University of Illinois Urbana–Champaign (UIUC)** IL, USA
Ph.D. in mathematics
Advisor: Prof. Jesse Thorner
Overall GPA: 4.00/4.00

2017–2021 **Bowdoin College** ME, USA
B.A., Mathematics major, Japanese minor, completed with honors in mathematics
Advisor: Prof. Naomi Tanabe
Overall GPA: 3.931/4.00, Magna Cum Laude

Semester abroad (Spring 2020) Budapest, Hungary
Budapest Semesters in Mathematics (BSM), completed with honors

RESEARCH INTEREST

Analytic number theory, Distribution of primes, L -functions, Automorphic forms.

PUBLICATIONS/PREPRINTS

1. *The $\theta = \infty$ Conjecture and the Riemann Hypothesis for Automorphic L -functions.*
(With Anji Dong and Alexandru Zaharescu, submitted 2026)(arXiv)
2. *Zeros of Polynomials in Derivatives of Automorphic L -functions.*
Mathematika 72, e70120 (2026). (arXiv, DOI)
(With Anji Dong and Alexandru Zaharescu)
3. *A metric approach to zero-free regions for L -functions.*
Eur. J. Math. 12, 30 (2026). (arXiv, DOI)
4. *Effective correlation and decorrelation for newforms, and weak subconvexity for L -functions.*
Math. Z. 313, 23 (2026). (arXiv, DOI)
(With an appendix by Jesse Thorner)
5. *Walking to infinity along some number theory sequences.*
Integers, 24 (2024), 29 pp. (arXiv, DOI)
(With Steven J. Miller, Fei Peng, Tudor Popescu, and the Polymath REU program)
6. *An alternative approach to computing $\beta(2k+1)$.*
Integers, 23 (2023), 13 pp. (arXiv, DOI)
(With Naomi Tanabe)
7. *Modeling random walks to infinity on primes in $\mathbb{Z}[\sqrt{2}]$.*
J. Integer Seq. 25 (2022), 21 pp. (arXiv, DOI)
(With Bencheng Li, Steven J. Miller, Tudor Popescu, and Daniel Sarnecki)
8. *Walking to infinity on the Fibonacci sequence.*
Fibonacci Quart. 60 (2022), no. 5, 293–299. (arXiv, DOI)
(With Bencheng Li, Steven J. Miller, Tudor Popescu, and Daniel Sarnecki)

TALKS ([†] = upcoming talks)**Invited Talks**

Feb 2026	Kansas State University Student Number Theory Seminar	Manhattan, KS
Jan 2026	Joint Mathematics Meetings (JMM) 2026: AMS Special Session on <i>Recent Trends in Analytic Number Theory</i>	Washington, DC
Dec 2025	PAImetto Number Theory Series (PANTS) 2025 <i>Invited graduate student speaker</i>	Clemson, SC
Jan 2025	Joint Mathematics Meetings (JMM) 2025: AMS Special Session on <i>Current Directions in Modular Forms</i>	Seattle, WA
Sep 2024	Kansas State University Student Number Theory Seminar	Remote
Jun 2024	Shandong University Number Theory Seminar	Remote
Apr 2024	Stanford University Student Analytic Number Theory Seminar	Remote
Mar 2024	UIUC Number Theory Seminar	Urbana, IL

Contributed Talks

Jun 2025	Annual Pure and Applied Mathematics Conference (APAM)	Bangkok, Thailand
Jun 2025	Conference in Number Theory and Applications (CNA)	Bangkok, Thailand
Apr 2025	Automorphic Forms Workshop (AFW)	Denton, TX
Apr 2025	UIUC Graduate Number Theory Seminar (Also in Oct 2024, Mar 2024, and Nov 2023)	Urbana, IL
Jul 2022	The 20 th International Conference on Fibonacci Numbers and Their Applications, University of Sarajevo, Bosnia and Herzegovina.	Remote
May 2021	Honors Thesis, Department of Mathematics, Bowdoin College	Remote
Sep 2020	The PAImetto Joint Arithmetic, Modularity, and Analysis Series (PAJAMAS)	Remote
Oct 2019	President's Summer Research Symposium, Bowdoin College	Brunswick, ME

WORKSHOPS & CONFERENCES ATTENDED ([†] = upcoming workshops/conferences)

Jun 2026	Women in Numbers 7, Banff International Research Station	Alberta, Canada
Jan 2026	Joint Mathematics Meetings (JMM) 2026	Washington, DC
Dec 2025	PAImetto Number Theory Series (PANTS) 2025	Clemson, SC
Jun 2025	Annual Pure and Applied Mathematics Conference (APAM)	Bangkok, Thailand
Jun 2025	Conference in Number Theory and Applications (CNA)	Bangkok, Thailand
Apr 2025	Automorphic Forms Workshop (AFW)	Denton, TX
Mar 2025	Midwest Arithmetic Geometry and Number Theory (MAGNTS)	Columbus, OH
Jan 2025	Joint Mathematics Meeting (JMM) 2025	Seattle, WA
May 2024	Shanks Conference, International Conference on <i>L</i> -Functions and Automorphic Forms	Nashville, TN
Oct 2023	Midwest Arithmetic Geometry and Number Theory (MAGNTS)	Ann Arbor, MI
Jan 2023	Spring Opportunities, IAS	Princeton, NJ
Jun 2022	The Connecticut Summer School in Number Theory (CTNT)	Storrs, CT
Jul 2019	Park City Mathematics Institute (PCMI)	Park City, UT

TEACHING EXPERIENCE

2026–Present	Lecturer , Department of Mathematics, UVA	Charlottesville, VA
	Fall 2026	MATH 3310 Basic Real Analysis
2021–2025	Teaching Assistant , Department of Mathematics, UIUC	Urbana, IL
	Fall 2025	MATH 231 Integral Calculus
	Summer 2024	MATH 257 Linear Algebra (NetMath)
	Spring 2024	MATH 220 Differential Calculus (Head TA)*
	Fall 2023	MATH 221 Differential Calculus*
	Spring 2023	MATH 220 Differential Calculus (Head TA)*
	Fall 2022	MATH 220 Differential Calculus*
	Spring 2022	MATH 220 Differential Calculus*
	Fall 2021	MATH 231 Integral Calculus*
	* = Ranked as an excellent teacher by students. There is no evaluation for NetMath and the ranking system has been retired since Fall 2025.	
	Grader , Department of Mathematics, UIUC	Urbana, IL
	Summer 2023	MATH 347 Fundamental Mathematics
2018–2019	Quantitative Reasoning Tutor , Bowdoin College	Brunswick, ME
	Supported students in courses such as mathematical reasoning, linear algebra, and calculus at dedicated study nights for students from underrepresented backgrounds.	
2018–2019	Grader , Bowdoin College	Brunswick, ME
	Fall 2019	MATH 2303 Functions of Complex Variables
	Fall 2019	MATH 2602 Introduction to Group Theory
	Fall 2018–Spring 2019	MATH 2020 Mathematical Reasoning

RESEARCH AND MENTORING EXPERIENCE

Fall 2024	Research Assistant	Urbana, IL
–Spring 2025	Supported by a Simons Foundation grant awarded to Prof. Thorner.	
Summer 2020	Research Mentor for the Polymath Jr. REU	Remote
	Guided undergraduate students through research projects and organized weekly activities under the supervision of Prof. Miller (Williams College), fostering early research skills.	
Summer 2018	Academic Program Assistant , Brewster Academy	Wolfeboro, NH
	Instructed Thai Government scholarship recipients in SAT test preparation.	
Winter 2018 & 2019	Academic Program Assistant , SIT Graduate institution	Brattleboro, VT
	Guided Thai Government scholarship recipients through U.S. college applications.	

OUTREACH

Fall 2024	Math Circle Instructor , UIUC	Urbana, IL
	Offered a free weekly extracurricular math program for middle school students, fostering enthusiasm and supporting younger learners in the Champaign–Urbana area.	
Summer 2022, 2017, & 2016	Mathematical Outreach Speaker , Benchama Maharat Highschool	Thailand
	Delivered lectures to over 200 middle and high school students, preparing them for national math competitions, expanding access to advanced mathematical preparation.	
Summer 2017	Community Teaching Volunteer , The Mirror Foundation	Chiangrai, Thailand
	Volunteered as a teacher for students from underrepresented ethnic groups, planning activities and providing short courses in various subjects.	

AWARDS

2026	Ruth V. Shaff and Genevie I. Andrews Fellowship , UIUC Awarded to a graduate student in Mathematics.	Urbana, IL
2026	Graduate Student Travel Grants to JMM 2026 , AMS Awarded a travel grant to present research at the 2026 Joint Mathematics Meetings (JMM).	Washington, DC
2025	Bateman Prize in Number Theory , UIUC Awarded to a graduate student in recognition of outstanding research in number theory.	Urbana, IL
2025	Paul T. Bateman Fellowship , UIUC Awarded to a student actively working on research and writing thesis in number theory.	Urbana, IL
2025	Departmental TA Instructional Award , UIUC Awarded to graduate teaching assistants for exemplary teaching.	Urbana, IL
2021	Edward S. Hammond Mathematics Prize , Bowdoin College Awarded to a senior who is completing mathematics major with distinction.	Brunswick, ME
2021	Phi Beta Kappa , Alpha of Maine Chapter	ME
2019–2021	Smyth Mathematics Prize , Bowdoin College Awarded to a student with highest mathematics grades during the first two years.	Brunswick, ME
2019	Kibbe Science Fellowship , Bowdoin College Fellowship for summer research in STEM in summer 2019.	Brunswick, ME
2017–2019	Sarah and James Bowdoin Scholar , Bowdoin College Given to the top 20 percent of students, as calculated by grade point average.	Brunswick, ME
2016–2021	King's Scholarship , Royal Thai Government A full-ride scholarship for overseas undergraduate studies.	Thailand

PROFESSIONAL SERVICES

Referee : International Mathematics Research Notices, Revue Roumaine des Sciences Techniques,
Mathematical Reports.